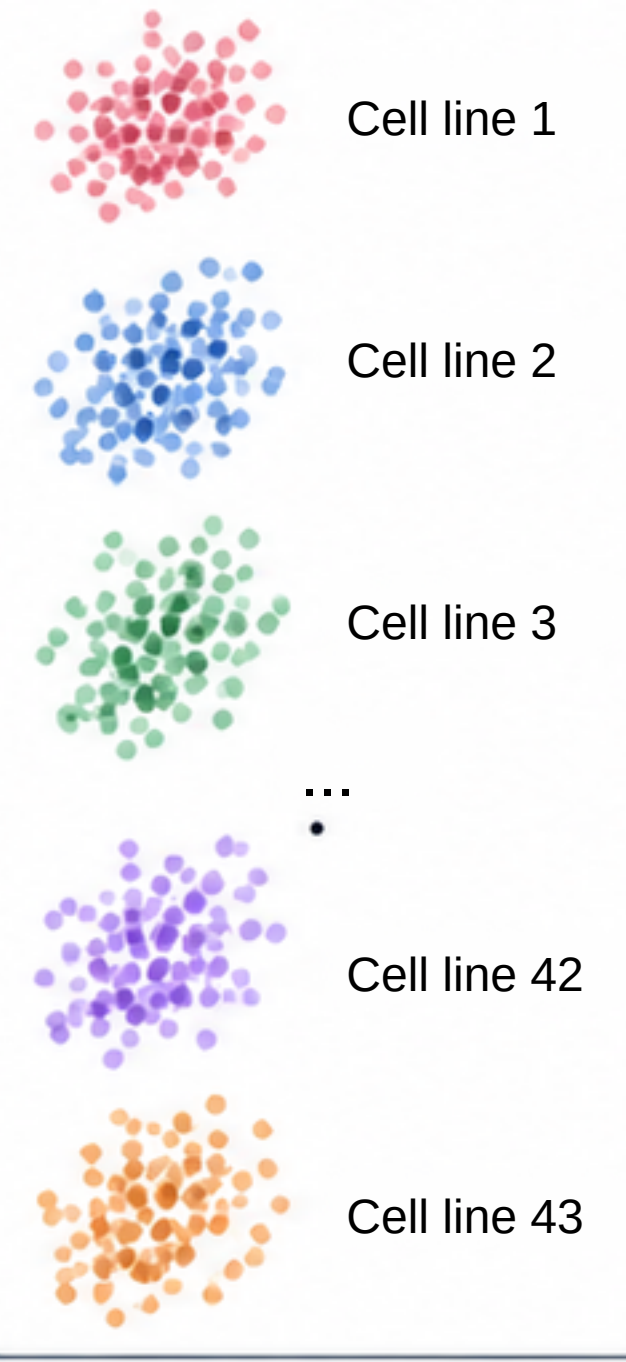


a

Training contexts

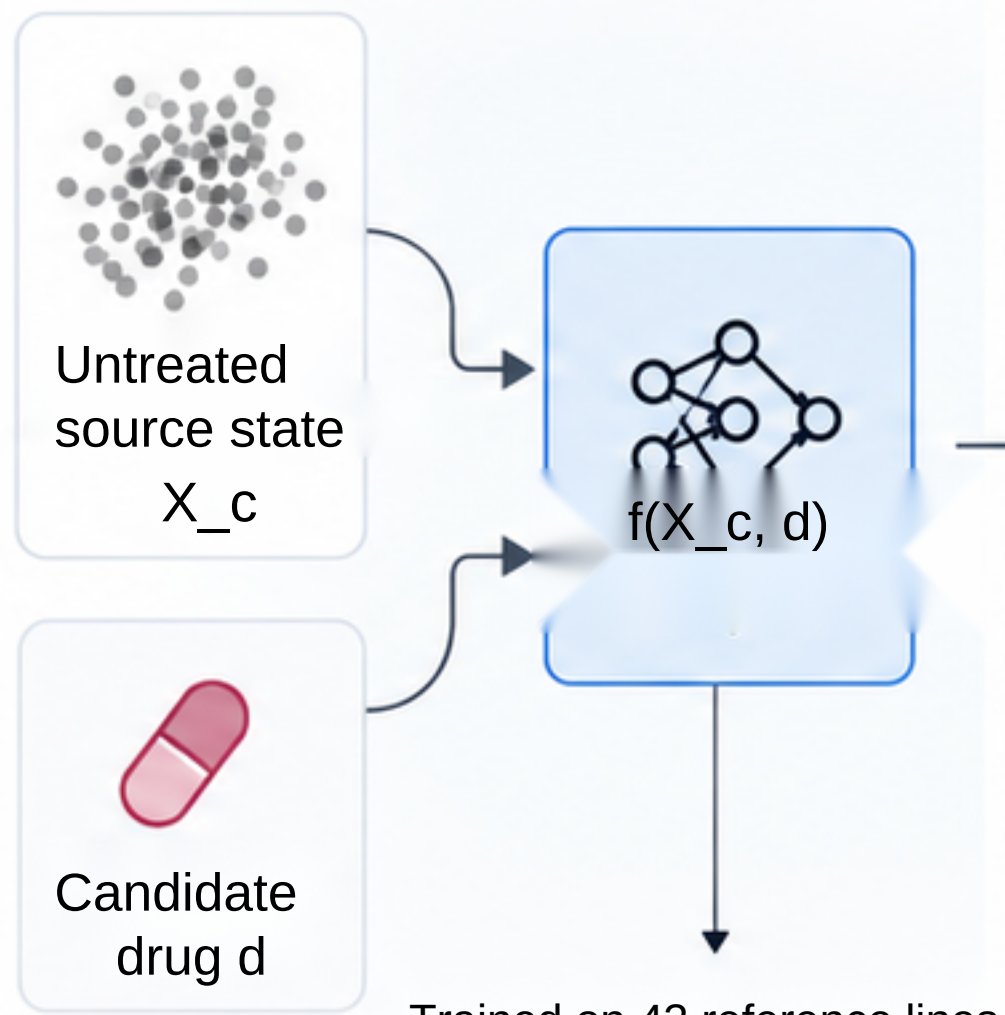
43 reference cell lines
observed drug responses



fit on 43 reference lines

Forward response predictor

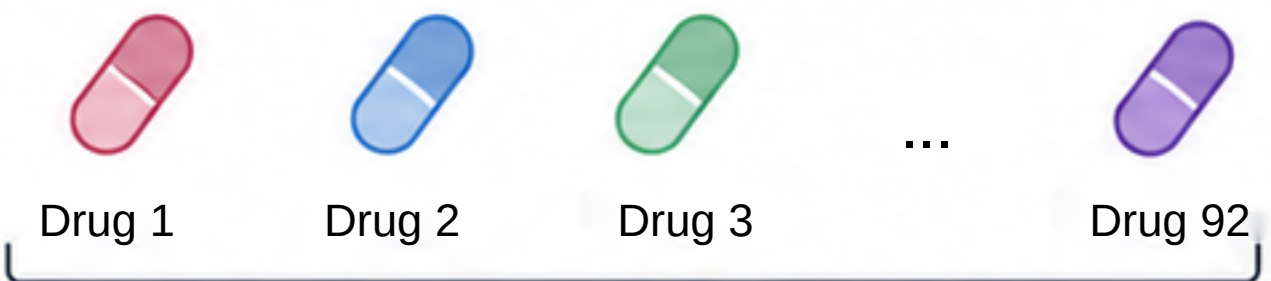
$f(\text{source state}, \text{drug})$



Trained on 43 reference lines
(held-out line excluded)

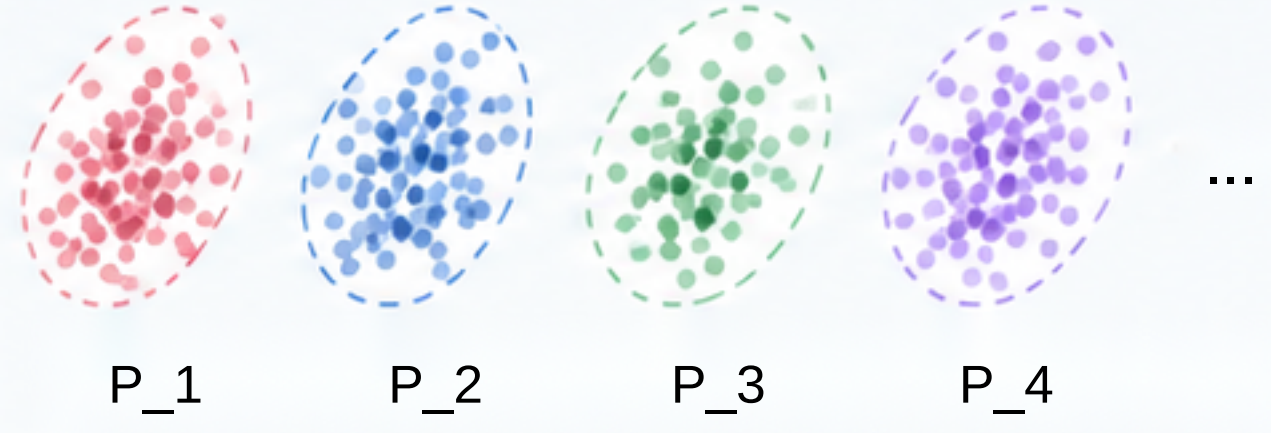
Held-out cell line

92 fixed candidate drugs



Predicted candidate responses

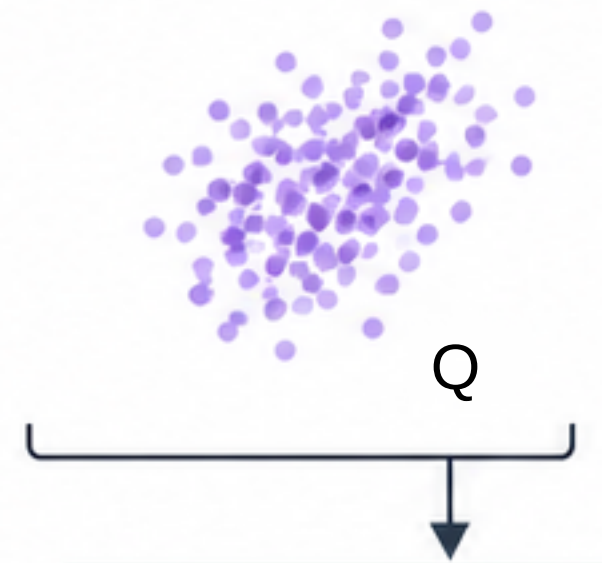
predicted in the held-out line, not observed



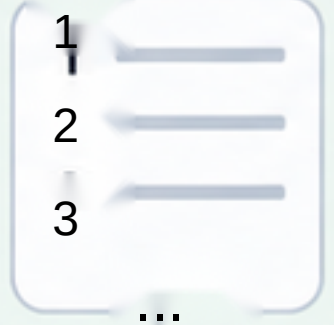
$P_{d,c} = f(X_c, d)$

Target population Q

Wanted state



Same target-matching and ranking procedure as Fig. 5a



Does the oracle population advantage survive prediction?

Prediction-constrained intervention-identification task in a held-out cell line

Candidate outcomes are predicted rather than observed; the target population is given only to the retrieval scorer.